AI SYSTEMS & BACKEND ENGINEERING CONSULTANCY

Production AI Systems · Scalable Backend Architecture · Cloud Infrastructure

Service Model: Technical Advisory & Implementation
Focus: Vibe Code → Production Software

Target Audience: CTOs, VPs of Engineering & Founders
Engagements: Audits, Refactoring & Deployments

Company Value Proposition: AI-assisted development allows teams to build prototypes faster than ever, but rapidly generated code often accumulates critical technical debt — duplicated logic, coupled components, vulnerable authentication, and unstable latency. **We help engineering leaders transform vibe-coded applications into clean, resilient, enterprise-ready production software** with strict data contracts, high-throughput model serving, and automated cloud CI/CD pipelines.

1. CORE CONSULTING & ENGINEERING SERVICES

Codebase Audit & Architecture Assessment Comprehensive review of existing applications to identify architecture flaws, security vulnerabilities, latency bottlenecks, and unmaintainable technical debt before it halts growth. **Key Deliverables:** Architecture Assessment Report, Auth Inspection, Refactoring Roadmap	**Refactoring & Domain Modularization** Transform tightly coupled or vibe-coded prototypes into clean, modular architectures with strict service boundaries and isolated domain logic. **Key Deliverables:** Domain Boundary Separation, Clean Schema Contracts, Decoupled Service Layers
Production Backend Engineering Architect and build robust, asynchronous FastAPI microservices, scalable database layers, and complex third-party API orchestrations engineered for real load. **Key Deliverables:** Asynchronous REST APIs, Relational/Vector Schemas, Third-Party Integration	**Applied AI & LLM Systems Engineering** Engineer production-ready LLM pipelines, dynamic contextual RAG frameworks, model fine-tuning (LoRA), and agentic workflows with strict validation. **Key Deliverables:** High-Throughput vLLM Serving, Dynamic RAG Pipelines, LoRA Fine-Tuning
Enterprise Productionization & Security Harden applications to meet enterprise compliance standards: Microsoft Entra ID RBAC, per-user OAuth 2.0 token handshakes, structured logging, and automated CI/CD. **Key Deliverables:** Enterprise OAuth 2.0 / RBAC, Azure CI/CD Pipelines, Audit Trail Systems	**Performance & Throughput Optimization** Diagnose and resolve GPU memory leaks, KV-cache fragmentation, concurrency limits, and p99 latency spikes across backend and AI workloads. **Key Deliverables:** GPU KV-Cache Tuning, Concurrency Stress Testing, Latency Profiling

2. FLAGSHIP CASE STUDIES & TECHNICAL DELIVERABLES

CASE STUDY 01: Enterprise LLM Proposal Engine [Client Deliverable]

Client Profile: Australian Engineering Consultancy | Measurable Impact: Turnaround reduced from Days → < 1 Hour

The Challenge: Client required an automated enterprise solution to transform project briefs into client-ready fee proposals with strict rate-card pricing and branded Word/Excel deliverables without risking hallucinated fee data or security audit gaps.

Consulting & Engineering Work Delivered:

- **FastAPI Architecture:** Built modular services on Azure App Service with isolated prompt-assembly logic.
- **Multi-Stage Generation:** Engineered context management preserving prompt fidelity across multi-step LLM workflows.
- **Zero-Audit-Gap Security:** Configured per-user OAuth 2.0 token handshakes with Total Synergy v4 API and Microsoft Entra ID RBAC across 24 mapped enterprise use cases, ensuring all mutations are tied to individual authenticated users.
- **CI/CD Pipeline:** Solved deployment timeouts and package limits using Azure DevOps Kudu async zipdeploy.

`Technologies: Python, FastAPI, Azure App Service, Azure DevOps, Microsoft Entra ID, OAuth 2.0, RBAC, REST APIs, LLMs`

CASE STUDY 02: High-Throughput LLM Inference Service **[AI Infrastructure]**

Target Environment: Single NVIDIA RTX 2080 Ti | Measurable Impact: 100 Concurrent Requests Supported

The Challenge: Deploying high-concurrency LLM inference on budget-constrained hardware without running into GPU VRAM out-of-memory errors or unacceptable queue waiting times.

Consulting & Engineering Work Delivered:

- **Continuous Batching & PagedAttention:** Integrated vLLM serving engine to eliminate memory fragmentation.
- **KV-Cache Optimization:** Tuned memory allocation chunks and configured asynchronous request schedulers.
- **Architecture Pipeline:** Client → API Gateway → Async Queue / Scheduler → vLLM Batcher → GPU VRAM → Streaming Tokens.
- **Synthetic Load Benchmarking:** Measured token-to-first-token (TTFT) and inter-token latency to maintain sub-second SLA.

`Technologies: Python, vLLM, Continuous Batching, PagedAttention, PyTorch, FastAPI, Docker, GPU Profiling`

CASE STUDY 03: Contextual Dynamic RAG Framework **[Search & Retrieval Infrastructure]**

Retrieval Architecture: Dynamic Multi-Strategy Routing | Measurable Impact: +60% Retrieval Efficiency Gain

The Challenge: Single-path naive RAG pipelines created unnecessary latency on exact-keyword searches and poor recall on multi-step reasoning queries across complex enterprise document sets.

Consulting & Engineering Work Delivered:

- **Dynamic Query Classifier:** Upfront classifier analyzing user intent, keyword density, and semantic complexity.
- **Hybrid Retrieval Layer:** Routes between FAISS dense vector search and BM25 sparse keyword indices dynamically.
- **Cross-Encoder Reranking:** Applied fine-grained reranking over candidate chunks prior to context window assembly.
- **Context Compaction:** Deduplicated and compacted retrieved passages to prevent prompt bloating and LLM distraction.

`Technologies: FAISS, BM25, Cross-Encoder, LangGraph, Python, FastAPI, Hugging Face Embeddings`

3. TECHNICAL STACK & CORE CAPABILITIES

Backend & Cloud	FastAPI, REST APIs, Asynchronous Python, Docker, Git, Azure App Service, Azure DevOps, Microsoft Entra ID, OAuth 2.0, RBAC, CI/CD Automation
AI / LLM Systems	LoRA / PEFT, Unsloth, vLLM, llama.cpp, GGUF Quantization, Dynamic RAG, Prompt Engineering, LangChain, LangGraph, Hugging Face Transformers, OpenAI / Anthropic / Gemini APIs
ML / Vision / Audio	PyTorch, TensorFlow, TensorFlow Lite, Ultralytics YOLOv8, OpenCV, Whisper (Speech-to-Text), scikit-learn, Optuna, Weights & Biases, Pandas, NumPy
Data & Storage	PostgreSQL, MySQL, MongoDB, Supabase, FAISS Vector Databases, Power BI, Tableau

4. CLIENT ENGAGEMENT HISTORY & TRACK RECORD

Enterprise AI & Proposal Systems **Australian Engineering Firm** **2026 — Present**	Delivered automated proposal generation architecture, multi-stage context assembly layer, Microsoft Entra ID RBAC across 24 use cases, and per-user OAuth 2.0 token handshakes with Total Synergy v4 API on Azure App Service.
AI Infrastructure & Model Serving **AI Engineering Consultancy** **2025 — 2026**	Architected high-throughput vLLM inference services (100 concurrent requests on RTX 2080 Ti), dynamic RAG frameworks (+60% retrieval efficiency), edge vision deployments (TF Lite GPU), and offline Android GGUF RAG chatbots.
Analytics & Applied Machine Learning **FinTech & Healthcare Sector** **2023 — 2024**	Built real-time financial price prediction models for banking decision support, digital receipt OCR extraction pipelines (+15% extraction precision), and executive data analytics dashboards.